

1. What will be the output of the following code?

```
import numpy as np  
  
arr = np.array([1, 2, 3, 4, 5])  
  
print(arr[1:4])
```

- a) [1, 2, 3]
- b) [2, 3, 4]
- c) [1, 2, 3, 4]
- d) [3, 4, 5]

Answer: b) [2, 3, 4]

2. Which of the following libraries is used for training neural networks in Python?

- a) Matplotlib
- b) TensorFlow
- c) Scikit-learn
- d) OpenCV

Answer: b) TensorFlow

3. In Keras, which of the following is used to add a dense layer with 128 units and 'relu' activation function?

- a) model.add(Dense(128, activation='relu'))
- b) model.add(Dense(128, input_dim=128, activation='relu'))
- c) model.add(Dense(128, units=128, activation='relu'))
- d) model.add(Dense(128, units=64, activation='relu'))

Answer: a) model.add(Dense(128, activation='relu'))

4. Which of the following code snippets is used to initialize a random forest classifier in scikit-learn?

- a) model = RandomForestClassifier(n_estimators=100)
- b) model = RandomForest(n_estimators=100)
- c) model = RandomForestClassifier()
- d) model = RandomForestModel(n_estimators=100)

Answer: a) model = RandomForestClassifier(n_estimators=100)

5. What will the following code print?

```
import numpy as np  
  
arr = np.array([1, 2, 3, 4])  
  
arr = arr * 2  
  
print(arr)
```

- a) [1, 2, 3, 4]
- b) [2, 4, 6, 8]
- c) [0, 1, 2, 3]
- d) [2, 3, 4, 5]

Answer: b) [2, 4, 6, 8]

6. What is the output of the following code using pandas?

```
import pandas as pd  
  
df = pd.DataFrame({'A': [1, 2, 3], 'B': [4, 5, 6]})  
  
print(df['A'] + df['B'])
```

- a) [5, 7, 9]
- b) [1, 2, 3]
- c) [4, 6, 8]
- d) Error

Answer: a) [5, 7, 9]

7. Which of the following will be the correct syntax to split the data into training and testing sets in scikit-learn?

- a) train_test_split(X, y, test_size=0.2)
- b) train_split(X, y, test_size=0.2)
- c) train_test_split(X, test_size=0.2)
- d) split_data(X, y, test_size=0.2)

Answer: a) train_test_split(X, y, test_size=0.2)

8. Which of the following code snippets is used to load a dataset from CSV in Python using pandas?

- a) `data = pd.load_csv('dataset.csv')`
- b) `data = pd.read('dataset.csv')`
- c) `data = pd.read_csv('dataset.csv')`
- d) `data = pd.read_csv_file('dataset.csv')`

Answer: c) `data = pd.read_csv('dataset.csv')`

9. In the following code, which function trains a model in scikit-learn?

```
from sklearn.linear_model import LogisticRegression  
  
model = LogisticRegression()  
  
model.train(X_train, y_train)
```

- a) `fit()`
- b) `train()`
- c) `learn()`
- d) `fit_model()`

Answer: a) `fit()`

10. What will the following NumPy code print?

```
import numpy as np  
  
arr = np.arange(10)  
  
print(arr[:3])
```

- a) `[0, 3, 6, 9]`
- b) `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]`
- c) `[0, 3, 6]`
- d) `[0, 1, 2]`

Answer: a) `[0, 3, 6, 9]`

11. What does the following code do in TensorFlow/Keras?

```
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
```

- a) It trains the model with the Adam optimizer and binary cross-entropy loss function.
- b) It compiles the model with the Adam optimizer and calculates accuracy.
- c) It creates the model with the Adam optimizer.
- d) It initializes the model using binary cross-entropy loss.

Answer: b) It compiles the model with the Adam optimizer and calculates accuracy.

12. In machine learning, which function is used to evaluate the performance of a model on test data in scikit-learn?

- a) `model.evaluate(X_test, y_test)`
- b) `model.predict(X_test)`
- c) `model.score(X_test, y_test)`
- d) `model.fit(X_test, y_test)`

Answer: c) `model.score(X_test, y_test)`

13. What will the following code output?

```
import pandas as pd

df = pd.DataFrame({'Name': ['John', 'Alice', 'Bob'], 'Age': [23, 34, 45]})

print(df['Age'].mean())
```

- a) 34.0
- b) 25.0
- c) 30.0
- d) 23.0

Answer: a) 34.0

14. What is the correct way to normalize a feature column in scikit-learn?

- a) `from sklearn.preprocessing import normalize`
- b) `from sklearn.preprocessing import StandardScaler`
- c) `from sklearn.preprocessing import MinMaxScaler`
- d) `from sklearn.preprocessing import PolynomialFeatures`

Answer: c) `from sklearn.preprocessing import MinMaxScaler`

15. Which method is used to get predictions from a trained machine learning model in scikit-learn?

- a) model.train(X_test)
- b) model.evaluate(X_test)
- c) model.predict(X_test)
- d) model.predict_output(X_test)

Answer: c) model.predict(X_test)

16. What is the output of the following Python code?

```
import numpy as np  
  
arr = np.random.rand(3, 3)  
  
print(arr.sum())
```

- a) The sum of all elements in the array
- b) A random number
- c) The shape of the array
- d) The mean of the array

Answer: a) The sum of all elements in the array

17. What will be the output of this code?

```
import numpy as np  
  
arr = np.array([5, 6, 7, 8])  
  
print(arr[-2:])
```

- a) [7, 8]
- b) [5, 6]
- c) [6, 7]
- d) [7]

Answer: a) [7, 8]

18. What is the correct code to import and use the Random Forest Regressor from scikit-learn?

- a) from sklearn.ensemble import RandomForestRegressor
- b) from sklearn.tree import RandomForestRegressor

- c) from sklearn.ensemble import RandomForestClassifier
- d) from sklearn.ensemble import RandomForest

Answer: a) from sklearn.ensemble import RandomForestRegressor

19. In scikit-learn, what method is used to split the data into training and test sets?

- a) train_test_split(X_train, y_train, test_size=0.3)
- b) train_test_split(X, y)
- c) train_split(X, y)
- d) split_data(X, y)

Answer: b) train_test_split(X, y)

20. What will the following code output?

```
import numpy as np  
  
arr = np.array([1, 2, 3, 4, 5])  
  
arr = arr[arr > 2]  
  
print(arr)
```

- a) [1, 2]
- b) [3, 4, 5]
- c) [1, 2, 3, 4, 5]
- d) [4, 5]

Answer: b) [3, 4, 5]

21. What will be the output of the following Python code?

```
import numpy as np  
  
arr = np.array([1, 2, 3, 4, 5])  
  
print(arr[:2])
```

- a) [1, 2, 3, 4, 5]
- b) [1, 3, 5]
- c) [1, 2, 3]
- d) Error

Answer: b) [1, 3, 5]

22. Which of the following Python libraries is used for deep learning tasks?

- a) Pandas
- b) TensorFlow
- c) Scikit-learn
- d) Seaborn

Answer: b) TensorFlow

23. Which of the following code snippets correctly initializes a simple feedforward neural network in TensorFlow/Keras?

a) `model = Sequential()`
`model.add(Dense(64, input_dim=8, activation='relu'))`
`model.add(Dense(1, activation='sigmoid'))`

b) `model = Sequential()`
`model.add(Dense(64, input_dim=8))`
`model.add(Dense(1, activation='softmax'))`

c) `model = Sequential()`
`model.add(Dense(8, input_dim=64, activation='relu'))`
`model.add(Dense(1, activation='sigmoid'))`

d) `model = Sequential()`
`model.add(Dense(128, input_dim=8, activation='relu'))`
`model.add(Dense(2, activation='relu'))`

Answer: a) `model = Sequential()`
`model.add(Dense(64, input_dim=8, activation='relu'))`
`model.add(Dense(1, activation='sigmoid'))`

24. In the following code, what does the function fit() do in the context of machine learning?

```
from sklearn.linear_model import LogisticRegression  
  
model = LogisticRegression()  
  
model.fit(X_train, y_train)
```

- a) It trains the model on the provided training data.
- b) It predicts the target variable for the given input data.
- c) It splits the data into training and testing sets.
- d) It tunes the hyperparameters of the model.

Answer: a) It trains the model on the provided training data.

25. What is the purpose of the dropout layer in a neural network model?

- a) To increase the size of the input layer
- b) To prevent overfitting by randomly setting a fraction of input units to 0 during training
- c) To speed up the training process
- d) To increase the number of epochs

Answer: b) To prevent overfitting by randomly setting a fraction of input units to 0 during training

26. What is the correct way to calculate the accuracy of a machine learning model in scikit-learn?

- a) model.evaluate(X_test, y_test)
- b) accuracy = model.score(X_test, y_test)
- c) accuracy = model.predict(X_test, y_test)
- d) accuracy = model.predict_accuracy(X_test, y_test)

Answer: b) accuracy = model.score(X_test, y_test)

27. What will the following Python code using NumPy output?

```
import numpy as np  
  
arr = np.array([[1, 2, 3], [4, 5, 6]])  
  
print(np.reshape(arr, (3, 2)))
```


- a) [[1, 2], [3, 4], [5, 6]]
- b) [[1, 2, 3], [4, 5, 6]]
- c) [[1, 2], [4, 5], [3, 6]]
- d) Error

Answer: a) [[1, 2], [3, 4], [5, 6]]

28. Which of the following code snippets correctly loads a dataset in Pandas for machine learning?

a)import pandas as pd

```
df = pd.read_csv('dataset.csv')
```

b)import numpy as pd

```
df = pd.load_csv('dataset.csv')
```

c)import pandas as data

```
df = data.read_csv('dataset.csv')
```

d)import pandas

```
df = pandas.import_csv('dataset.csv')
```

Answer: a) import pandas as pd

```
df = pd.read_csv('dataset.csv')
```

``

29.What will be the output of the following Python code?

```
x = 5
```

```
y = 3
```

```
print(x + y)
```

- a) 53
- b) 8

- c) 5 + 3
- d) Error

Answer: b) 8

30. Which of the following is used to define a function in Python?

- a) function
- b) def
- c) func
- d) lambda

Answer: b) def

31. What is the correct syntax for a Python for loop?

- a) for i in range(10)
- b) for i = 0; i < 10; i++
- c) for (i in range(10))
- d) for i: range(10)

Answer: a) for i in range(10)

32. What will be the result of this Python expression?

```
x = "hello"
```

```
print(x[1])
```

- a) h
- b) e
- c) l
- d) o

Answer: b) e

33. Which of the following is the correct way to create a list in Python?

- a) list = ()
- b) list = []
- c) list = {}
- d) list = ;

Answer: b) list = []

34.What will be the output of the following code?

```
def add(x, y=10):
```

```
    return x + y
```

```
print(add(5))
```

- a) 15
- b) 5
- c) 10
- d) Error

Answer: a) 15

35.What will be the output of the following code?

```
x = [1, 2, 3]
```

```
x.append([4, 5])
```

```
print(len(x))
```

- a) 5
- b) 4
- c) 3
- d) Error

Answer: b) 4

36.Which method is used to add an element to the end of a list in Python?

- a) insert()
- b) append()
- c) extend()
- d) push()

Answer: b) append()

37.What does the following code do?

```
x = {1: "one", 2: "two", 3: "three"}
```

```
x.pop(2)
```

`print(x)`

- a) Removes key 1
- b) Removes key 2
- c) Removes the value of key 2
- d) Error

Answer: b) Removes key 2

38.Which of the following Python keywords is used for exception handling?

- a) throw
- b) try
- c) catch
- d) error

Answer: b) try

39.What is the output of the following code?

```
x = "Python"
print(x[:4])
```

- a) Python
- b) Pyth
- c) Pyt
- d) Error

Answer: b) Pyth

40.What is the purpose of the self keyword in a Python class?

- a) It refers to the current module.
- b) It refers to the parent class.
- c) It refers to the current instance of the class.
- d) It is used for inheritance.

Answer: c) It refers to the current instance of the class.

41.Which of the following is the correct syntax for a Python while loop?

- a) while x > 5 {
- b) while x > 5:
- c) while x > 5 do:
- d) while x > 5

Answer: b) while x > 5:

42.What will be the output of the following code?

```
x = [1, 2, 3]
```

```
y = x
```

```
y[0] = 10
```

```
print(x)
```

- a) [1, 2, 3]
- b) [10, 2, 3]
- c) [1, 10, 3]
- d) Error

Answer: b) [10, 2, 3]

43.What is the output of the following code?

```
def func(a, b=2):
```

```
    return a * b
```

```
print(func(3))
```

- a) 6
- b) 3
- c) 2
- d) Error

Answer: a) 6

44.Which method is used to merge two dictionaries in Python 3.9+?

- a) merge()
- b) combine()
- c) update()
- d) | (Pipe Operator)

Answer: d) | (Pipe Operator)

45.What will be the output of this code?

```
x = {1, 2, 3}
```

```
x.add(4)
```

```
print(x)
```

a) {1, 2, 3, 4}

b) {1, 2, 3}

c) {4, 1, 2, 3}

d) Error

Answer: a) {1, 2, 3, 4}

46. What is the output of the following code?

```
x = [1, 2, 3, 4]
```

```
y = x[::-1]
```

```
print(y)
```

a) [1, 2, 3, 4]

b) [4, 3, 2, 1]

c) [1, 3, 2, 4]

d) Error

Answer: b) [4, 3, 2, 1]

47.What is the output of the following code?

```
x = {1, 2, 3}
```

```
y = {3, 4, 5}
```

```
print(x & y)
```

a) {1, 2, 3, 4, 5}

b) {1, 2}

c) {3}

d) Error

Answer: c) {3}

48.What is the purpose of the `__init__()` method in a Python class?

- a) It initializes the class variables.
- b) It initializes the instance of the class.
- c) It initializes the class methods.
- d) It is called before any other methods.

Answer: b) It initializes the instance of the class.

49.Which of the following is used to create an anonymous function in Python?

- a) lambda
- b) def
- c) function
- d) anonymous

Answer: a) lambda

50.What is the output of the following code?

```
x = [1, 2, 3]
```

```
y = x
```

```
x = x + [4]
```

```
print(y)
```

- a) [1, 2, 3, 4]
- b) [1, 2, 3]
- c) [4, 1, 2, 3]
- d) Error

Answer: b) [1, 2, 3]